

Chorema

Geospatial future intelligence

Product overview

Andrea Modica | Martino Bellincampi

Rome & Copenhagen | 17 June 2026

The challenge We address

Public and private organizations are facing unprecedented volumes of complex **geospatial data without having adequate tools at their disposal.**

The lack of integrated predictive models limits the ability to anticipate future scenarios. Existing solutions (traditional GIS) provide only historical analysis and static visualizations, without predictive capabilities.

Problems and key questions

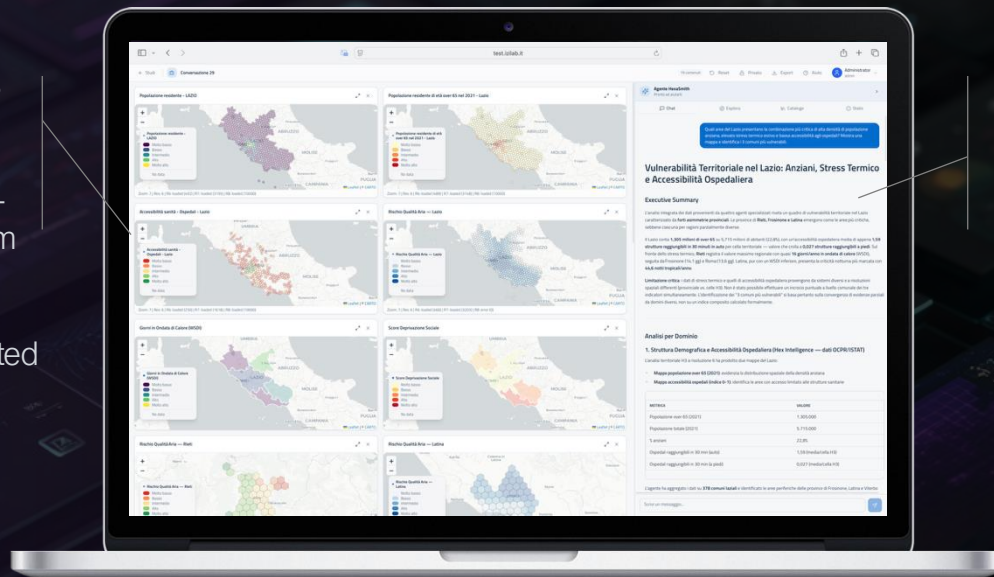
Problems we address:

- Strategic decisions made under deep uncertainty and rapid change
- Fragmented signals that fail to translate into actionable insight
- Over-reliance on short-term planning and linear forecasts
- Lack of internal foresight capability and shared sense-making
- Difficulty aligning strategy, policy, and investment to long-term goals

Typical questions we help answer:

- How can cities develop policies that make them more resilient to climate change, demographic shifts, and migration?
- Which emerging trends and weak signals will most likely disrupt specific market sectors in the next 5–15 years?
- How robust is a company's current strategy across different plausible futures?
- Where should we invest today to remain competitive and relevant tomorrow?
- What early indicators should we monitor to anticipate systemic risks or opportunities?

Complex spatial dynamics are analyzed through bespoke and scientifically validated indices (e.g., 15-minute city indices, tourism resilience indices, climate migration indices, climate risk indices, etc.) and plotted into maps



The user interacts with a cohort of thematically specialized AI Agents that constantly scan authoritative global data sources (UN, Eurostat, Copernicus, etc.) and scientific outlets (conferences, journals), ingest new information into the Chorema Data Lake and process it to derive updated geospatial knowledge

An easy-to-use multimodal and cross-media interface lets users save their projects and export reports in multiple formats. In terms of data ontology, the platform can be easily configured to let Clients import and process their proprietary data.

How is Chorema different from common GenAI platforms?

Let's look a case study

Foreskill is a labour market intelligence initiative developed through Chorema and integrating employment data, skills analysis, migration corridors, and diaspora dynamics with forward-looking scenarios to 2035, supporting policymakers on labour mobility and skills governance.

Same question.

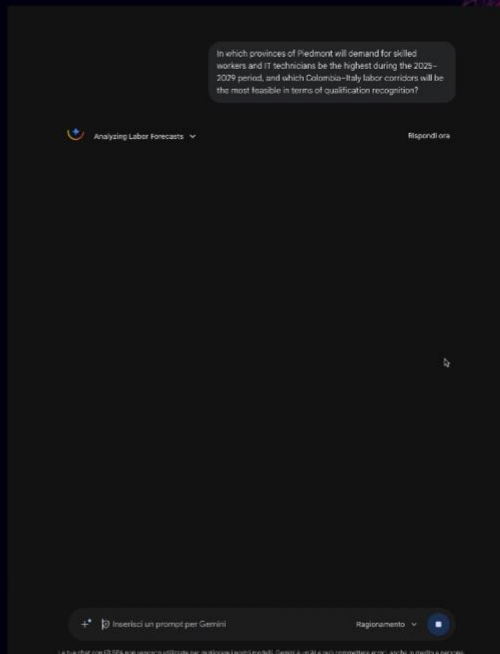
"In which provinces of Piedmont will demand for skilled workers and IT technicians be the highest during 2025–2029, and which Colombia–Italy labor corridors will be the most feasible in terms of qualification recognition?"

In which provinces of Piedmont will demand for skilled workers and IT technicians be the highest during the 2025–2029 period, and which Colombia–Italy labor corridors will be the most feasible in terms of qualification recognition?

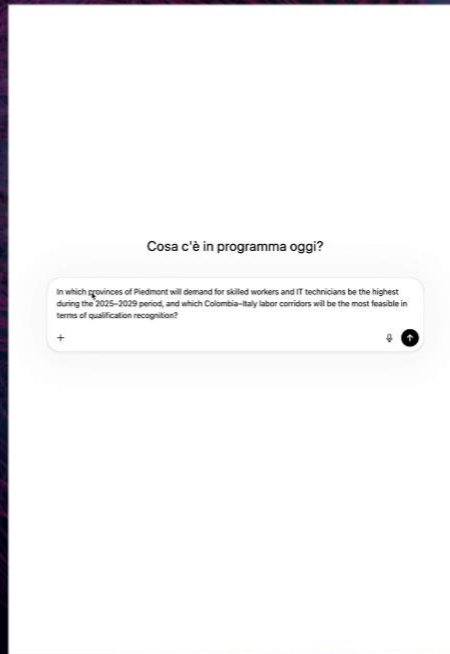


Same question. Three generalist answers.

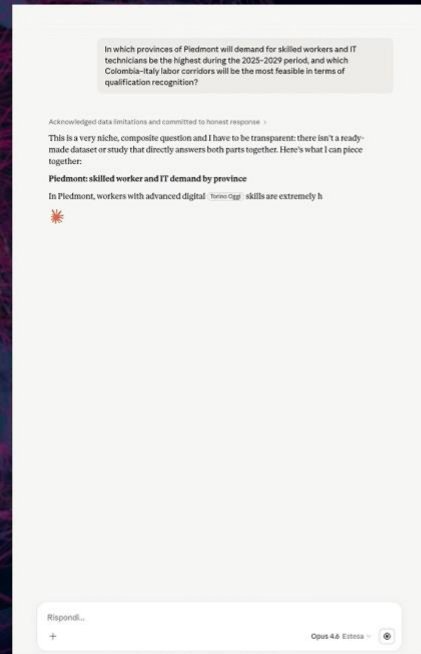
"In which provinces of Piedmont will demand for skilled workers and IT technicians be the highest during 2025–2029, and which Colombia–Italy labor corridors will be the most feasible in terms of qualification recognition?"



ChatGPT



Gemini



Claude



Same question. Three generalist answers.

"In which provinces of Piedmont will demand for skilled workers and IT technicians be the highest during 2025–2029, and which Colombia–Italy labor corridors will be the most feasible in terms of qualification recognition?"

Best generalist response.
Traced sources, explicit reasoning — but still a synthesis of secondary sources.

No real numbers.

Most structured — but cites specific corridors without explaining the selection logic.

**Why those?
On what basis?**

Transparent on limits — but no actionable output.

"I don't have this data, shall I search?"



ChatGPT



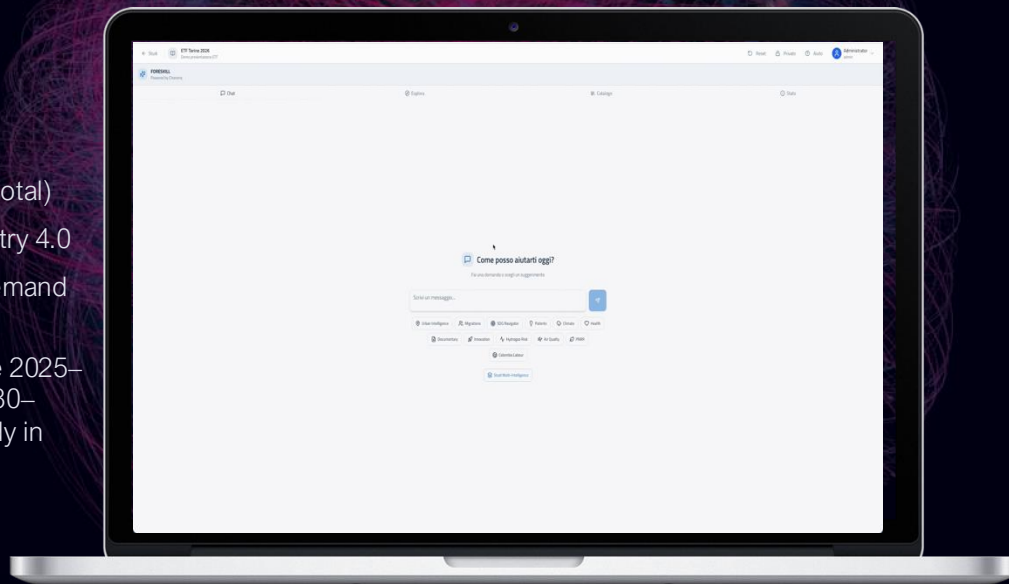
Claude

Same question. Curated knowledge.

"In which provinces of Piedmont will demand for skilled workers and IT technicians be the highest during 2025–2029, and which Colombia–Italy labor corridors will be the most feasible in terms of qualification recognition?"

ForeSkill

- **Turin:** 101,600 ICT jobs — 53,600 expansion demand (60% of regional total)
- **Cuneo:** strong manufacturing + Industry 4.0
- **Biella, Vercelli, Asti:** ICT expansion demand at zero
- **Corridors by time horizon:** Commerce 2025–27 | Manufacturing 2027–30 | ICT 2030–35~8,000 Colombian residents already in Piedmont





The structural difference

The difference is not in the model. It's in the knowledge infrastructure underneath.

	Claude	ChatGPT	Gemini	ForeSkill
Data Granularity	No	No	Partial	Yes
Expansion vs replacement	No	No	No	Yes
Negative intelligence	No	No	No	Yes
Corridor time horizons	No	No	No	Yes
Diaspora data	No	No	No	Yes
Traceable sources	Generic	Good	Uncertain	Curated (18 sources)
Hallucination risk	Low*	Low-medium	High	Contained
Actionability	None	Partial	Apparent	Full

Key features of Chorema

A platform that transforms complex geospatial data into actionable insights, integrating advanced predictive models with spatial analysis and agentic AI.

1

Thematic intelligence

8+ specialized domains (urban, climate, health, migration, hydrogeological risk, air quality, transport, biodiversity monitoring, and more) — each with dedicated AI agents and curated data.

2

Validated indices

These domains are analyzed through bespoke indices, i.e. descriptive and predictive algorithms that provide predefined analyses of complex geospatial dynamics. Indices are developed in collaboration with key international research institutions (e.g. Politecnico di Milano, Aalborg University, and LUM). New indices are created on demand via the Index Crafter module.

3

Agentic analysis

Multi-agent architecture — a coordinator agent interprets research questions, expert agents access domain-specific intelligence, a critic agent flags data gaps and hallucinations. Results grounded in the internal Data Lake, not generic LLM knowledge.

GEOSPATIAL AI 8 Intelligence Domains

Geospatial Future Intelligence

Esplora indici urbani, climatici, sanitari e di innovazione. Analisi geospaziale avanzata con griglie H3 e intelligenza artificiale.

[▶ Esplora](#)[📄 Scopri di più](#)[📄 6 Urban](#) [🌤️ 1 Climate](#) [💡 12 Innovation](#) [🌐 4 SDG](#) [🔬 1 Patents](#) [❤️ 5 Health](#)[🌊 6 Hydrogeo](#) [🌬️ 9 Air Quality](#)

📈 In Tendenza

Gli indici più esplorati questa settimana



Accessibilità Trasporti
OCPR sintetico mobilità pubblica e privata

[OCPR](#) [Trasporti](#)

Score Startup Combinato
Indice combinato ecosistema startup

[Startup](#) [Score](#)

Rischio Climatico Composito
Score composito 0-100 rischio climatico

[Score](#) [Composito](#)

Score Pericolosità Frane
Indice pericolosità da frana

[Frane](#) [Score](#)

Rischio Qualità Aria
Indice composito rischio qualità aria

[Composito](#) [Rischio](#)

Indice Dipendenza Maschi
Rapporto dipendenza demografica maschile

[Dipendenza](#) [Maschi](#)

📄 Urban Intelligence

Indici di accessibilità OCPR sintetici



How we build our indexes

Source indicators

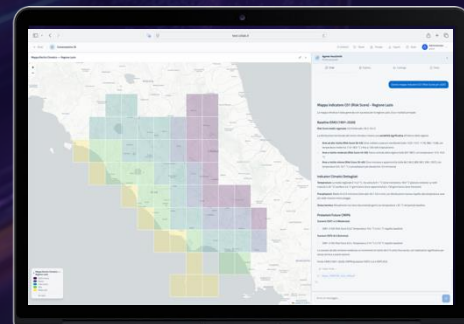
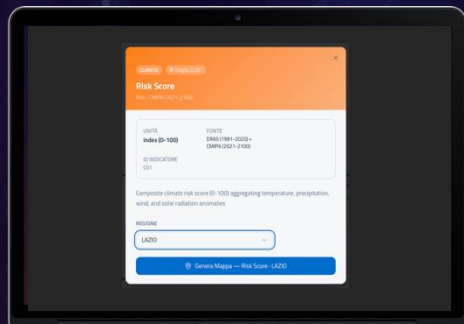
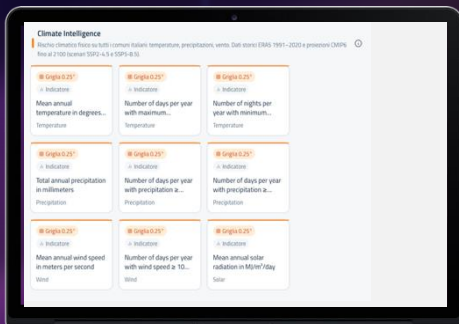
We collect and curate individual metrics from institutional, satellite, and real-time data sources — each measuring a single dimension of a territorial phenomenon.

Composite index design

Selected indicators are combined through weighted algorithms — co-developed with research partners — into a single interpretive or predictive index.

Territorial intelligence

The index is computed at the chosen geographic scale and rendered as an interactive map layer — ready for exploration, comparison, and scenario analysis.



How Chorema produces knowledge

01 Core knowledge production pipeline

INTERNAL DATA LAKE

- 80.000+ geospatial indicators, 500+ foresight and geopolitical reports
- ETL control center scanning hundreds of reputable sources (e.g. Copernicus, Corine Land Cover, etc.)
- Pre-processed and organized into thematic intelligence domains
- Real-time IoT/IloT sensor feeds ingested as live data streams

INDEX-BASED PROCESSING

- Indices: more than 100+ algorithms that combine multiple indicators to interpret or predict complex geospatial dynamics
- Created by: (1) internal analysts, (2) domain experts (Aalborg, PoliMi), (3) AI-powered Index Crafter scanning peer-reviewed sources (e.g., top scientific journals)

02 Additional knowledge production

EXTERNAL CONTEXTUAL KNOWLEDGE

- Chorema is connected to external LLMs (e.g., Claude) that provide additional and contextual knowledge that complements and integrates our core knowledge production pipeline

HALLUCINATION DETECTION

- Our bespoke AI critical agent controls and validates the contextual knowledge and assigns a level of reliability
- Our UI has a different color coding for information with lower levels of reliability so that the user can always know on which knowledge they can really count



Know what you can trust:
Each piece of information is tagged
by source and confidence level —
from verified institutional data to
AI-generated context.

● CHOREMA INTELLIGENCE — QUERY RESULT

QUERY

What is the hydrogeological risk exposure of the Liguria region, and how does it compare to national averages?

Liguria ranks among Italy's most exposed regions to hydrogeological risk. According to ISPRA data (2024), **91.1% of municipalities** are classified in high or very high risk zones — significantly above the national average of 78.3%. The region's composite vulnerability index (CVI) stands at **0.83**, driven by steep terrain, dense urbanization along narrow coastal strips, and aging drainage infrastructure.

Cross-referencing Copernicus land-cover dynamics (2018–2024) with precipitation anomaly trends, the Chorema **Territorial Stress Index** identifies three critical clusters: the Polcevera valley (Genova), the Cinque Terre coastal zone, and the Savona hinterland. All three show accelerating risk trajectories over the past 36 months.

Recent academic literature suggests that climate projections for the north-western Mediterranean basin point to an increase in short-duration, high-intensity rainfall events by 15–25% by 2050. If confirmed, this would significantly amplify existing vulnerabilities in Liguria's already fragile drainage systems.

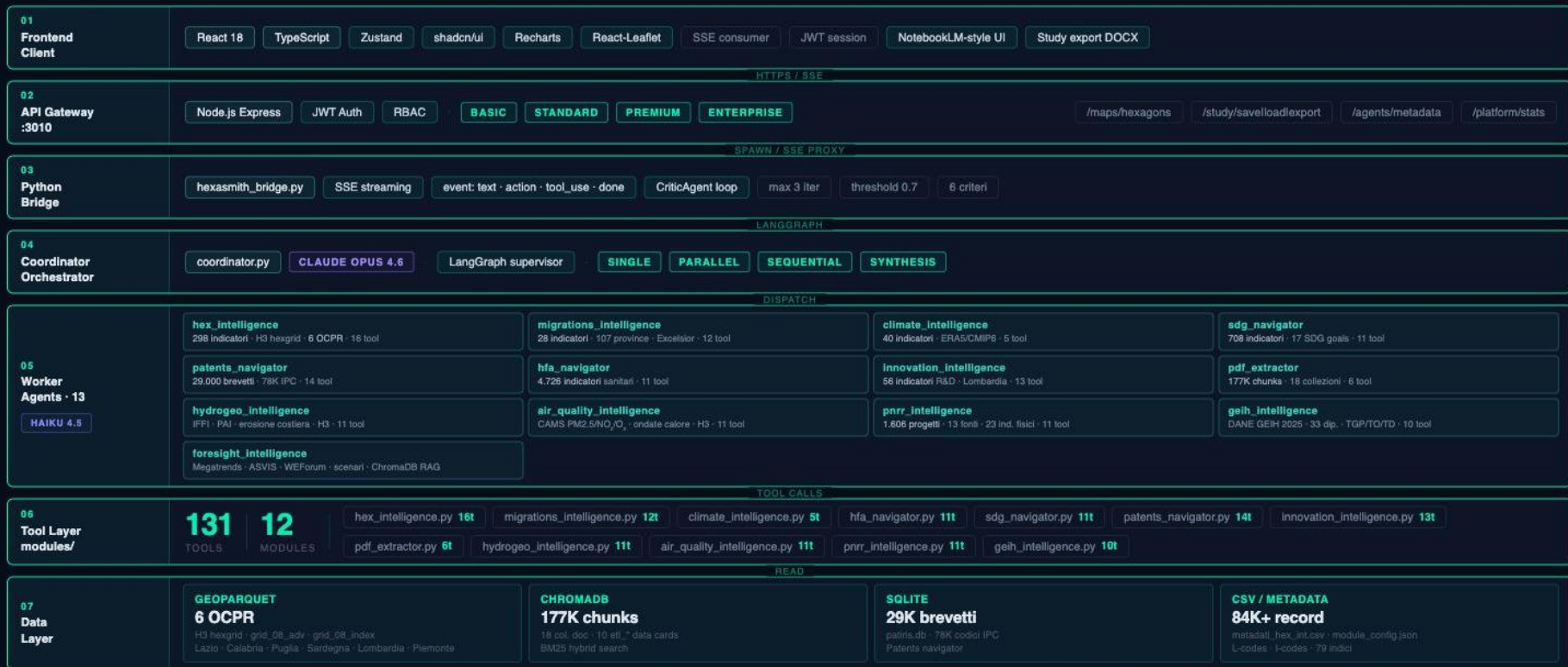
■ **Verified** — Data Lake & institutional sources

■ **Computed** — Chorema composite index

■ **Contextual** — LLM-sourced, lower confidence

Overall response confidence

78%



Chorema:

A distinctive approach to strategic planning and decision-making

01 Predictive analysis and future modeling:

Not just forecasting: scenario modeling, stress testing, and weak signal detection across 5–20 year horizons.

02 Agentic intelligence:

Custom-built AI agents that reason over structured territorial data, flag gaps, and cite sources — not generic chatbots.

03 Netflix-style discovery:

An editorial home page that lets non-expert users explore indices by theme, then seamlessly transition into a research studio for deeper analysis.

04 Data marketplace:

API-based access to curated territorial data streams — the “Netflix of big data” for external partners and developers.

User benefits: How Chorema addresses key needs

- 01 National institutions & ministries:**
Strategic intelligence on infrastructure portfolios, investment monitoring, policy impact assessment.
- 02 Cities & regional governments:**
Climate adaptation, 15-minute city planning, cultural observatories, S3 monitoring.
- 03 Corporate & energy:**
Innovation foresight, competitive benchmarking, consumer trend anticipation, service design.
- 04 Defense & strategic planning:**
Wargaming scenarios, network analysis, resource inventorying, parametric simulation on territorial variables.

Competition & positioning

It is not a dashboard, it is a research environment.

Chorema differentiates itself from traditional GIS platforms (Esri, QGIS), location intelligence tools (CARTO, Targomo, Placer.ai), and consulting-driven analytics (Deloitte, McKinsey) by **combining long-term foresight with agentic AI reasoning on scientifically validated territorial indices** that work as predefined units of analysis.

Chorema in detail: The features that make the product stand out

- 01 Bespoke methodology for multidimensional quantitative & qualitative analysis (big data, alternative data, expert elicitation); indices developed in the past 5 years through collaborations with 10+ universities in Italy and Europe.
- 02 Multi-agent architecture with coordinator, expert, and critic agents to detect emerging signals, flag data gaps, and deliver grounded early indicators.
- 03 Integration of design and foresight methods — in line with state-of-the-art scientific practices.
- 04 A dedicated team focused on visual storytelling, narrative sense-making, and decision-focused communication.



Understanding territories today

From data fragmentation to integrated vision

Integrated vision

Aggregate heterogeneous data (demographic, climatic, economic, infrastructure) into a single coherent territorial representation.

Flexible ontology and multi-source ETL pipelines.

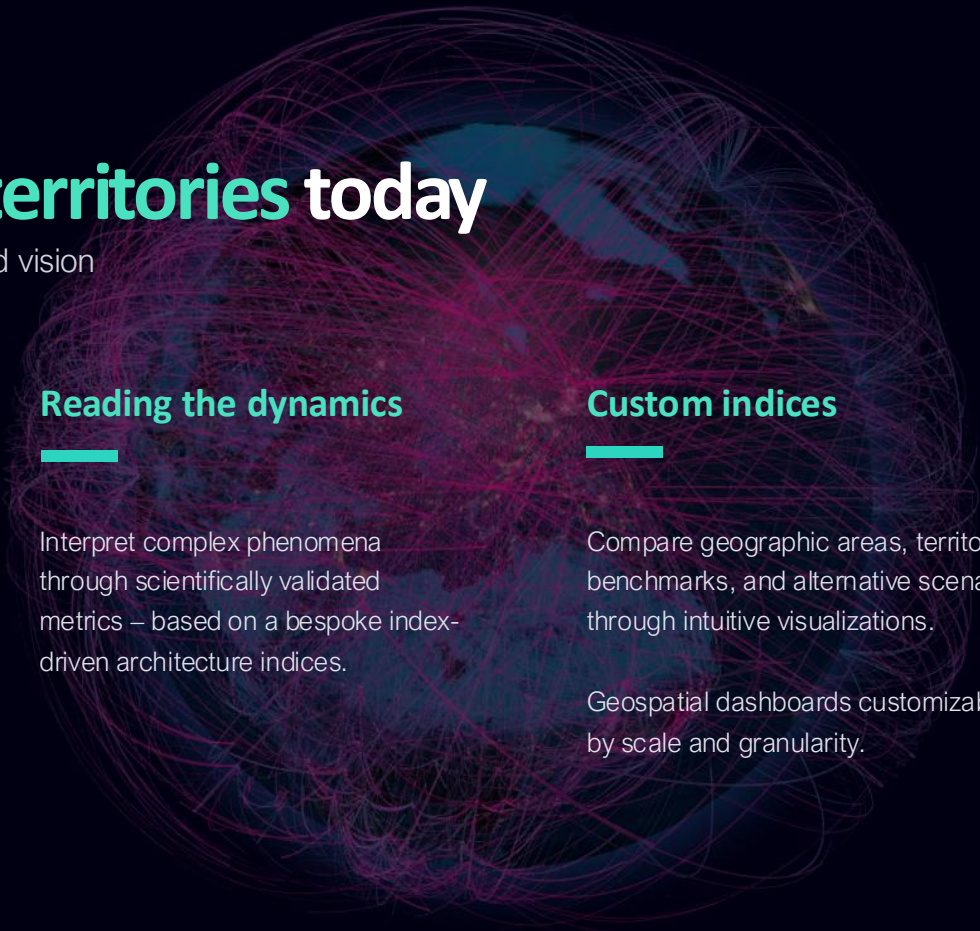
Reading the dynamics

Interpret complex phenomena through scientifically validated metrics – based on a bespoke index-driven architecture indices.

Custom indices

Compare geographic areas, territorial benchmarks, and alternative scenarios through intuitive visualizations.

Geospatial dashboards customizable by scale and granularity.



Anticipating the future

From today's snapshot to tomorrow's scenarios

Forecasting

Simulate territorial evolution over 5, 10, and 20 years across climatic, demographic, and economic dimensions.

Predictive layers with machine-learning models and historical time series.

Early warning

Detect weak signals before they become established trends: climate migration, shifts in consumption, urban tensions.

Horizon scanning + Composite Indices (Level 2): Resilience, Disruption, Emerging Potential

Future-proof strategies

Validate investments and policies against multiple scenarios, reducing the risk of obsolete decisions.

Scenario modeling and stress testing on key variables.





The Chorema ecosystem

One platform, multiple intelligence domains. Each vertical inherits the full Chorema stack — indices, agents, data lake — and adds domain-specific data and logic.

ForeSkill

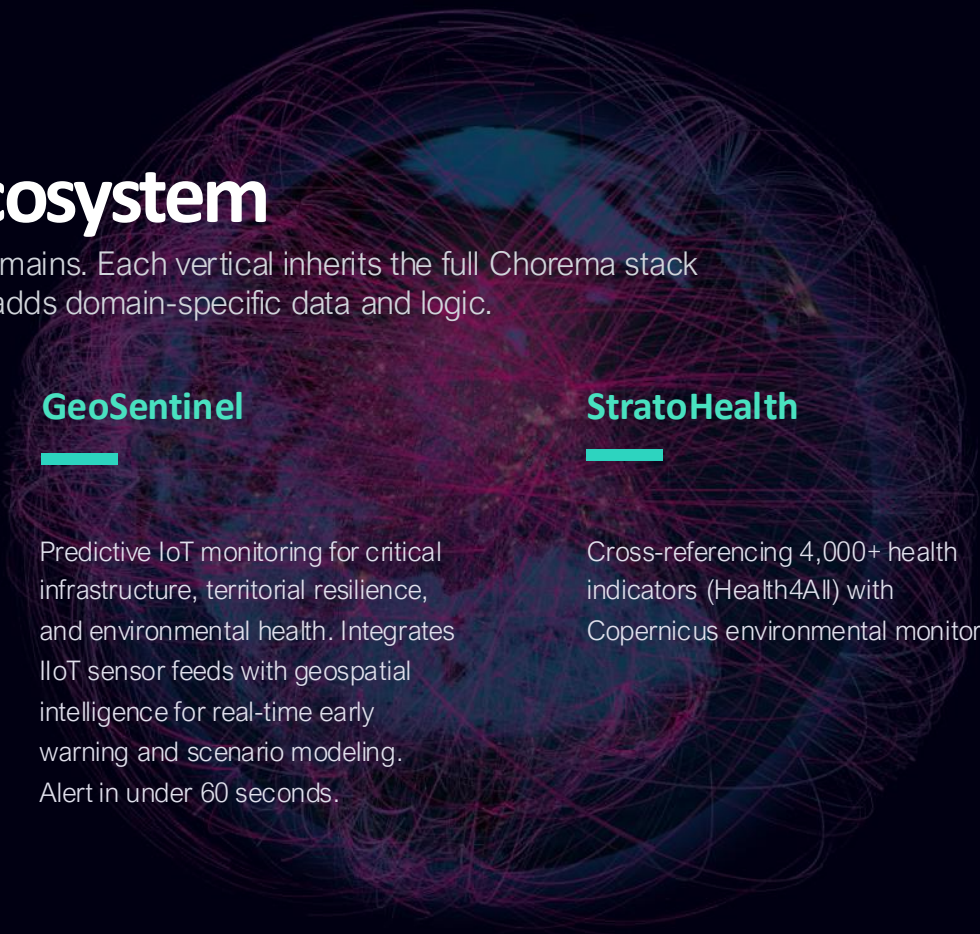
Europe's first SaaS for medium-term labour mismatch forecasting, applied to migration diasporas. Integrates CEDEFOP, Eurostat, Excelsior data.

GeoSentinel

Predictive IoT monitoring for critical infrastructure, territorial resilience, and environmental health. Integrates IIoT sensor feeds with geospatial intelligence for real-time early warning and scenario modeling. Alert in under 60 seconds.

StratoHealth

Cross-referencing 4,000+ health indicators (Health4All) with Copernicus environmental monitoring.



Chorema for Institutional Strategic Intelligence

User Stories – Ministry of Infrastructure and Transports

Who is the client

The Ministry of Infrastructure and Transport manages the largest PNRR portfolio in the transport sector: over 1,600 railway projects with a total value of €40 billion. An unprecedented data asset that requires advanced tools for visualization and strategic analysis.

Objectives

Integrate and normalize data from multiple sources into a single territorial intelligence platform; build a coherent taxonomy to classify 1,547 projects by type, scale, and impact; visualize the geographic distribution of €40 billion in investments through interactive geospatial dashboards; analyze differentiated impacts across Italy's 20 regions to support strategic monitoring and resource allocation decisions.

Problems & Challenges

With over 1,600 projects nationwide, the Ministry was managing fragmented data from multiple sources. The main challenges were aggregating information, analyzing the geography of investments, understanding impacts, and transforming data to support policy-making and infrastructure portfolio monitoring.

Chorema for Innovation Foresight

User Stories – The Case of an Energy Operator

Who is the client

A leading Italian energy operator engaged in the transition toward sustainable business models. The Innovation team required advanced tools to anticipate market transformations and guide product and service strategies in a sector characterized by rapid technological disruption and changing consumption patterns.

Objectives

Map emerging trends in energy consumption and market dynamics through systematic horizon scanning; build competitive benchmarks on the innovation trajectories of key players; identify patterns of technological and behavioral evolution to anticipate customer needs; support the service design of new offerings with a future-oriented analytical foundation.

Problems & Challenges

The sector faces a convergence of transformative forces: decarbonization, electrification, and new consumption behaviors. Key challenges include identifying weak signals of change ahead of competitors, the absence of benchmarks on sector innovation trajectories, and the need to translate forward-looking insights into truly future-proof services and products.



Trusted by

National institutions, energy leaders, cultural organizations, international partners, and research consortia rely on Chorema for strategic intelligence.

National institution



Energy



City government



Agrifood consortium



Cultural services



International migration





From pilot to ecosystem

From pilot project to data ecosystem

On-demand access

Consume territorial data streams without internal development, with flexible pay-per-use pricing.

Marketplace APIs: the “Netflix of territorial big data.”

Tailor-made indices

Co-create sector-specific indicators with academic methodological validation.

Index Crafter + partnerships with universities and research centers.

Enterprise integration

Embed Chorema into existing enterprise systems for a unified workflow.

REST APIs, standard connectors, multi-format export.

Pricing model: KaaS annual subscription + metered API calls for agentic queries + co-design services for index development.

Pricing & purchase options

Three commercial tiers mapped to platform access levels — plus flexible API and co-design.

MOST POPULAR

Starter

Municipalities · DMOs · small agencies

€20k / 6 months

- 1 intelligence domain
- Curated, validated indices
- Core agentic queries (metered)
- Studio + dashboards, link sharing
- Email onboarding & support

Access · Basic / Standard

Professional

Regions · mid-size firms · infra operators

€40–60k / year

- Up to 4 domains, custom index set
- Full multi-agent (coordinator + critic)
- Marketplace API access included
- Higher query volume, multi-format export
- Priority support + quarterly review

Access · Premium

Enterprise

Ministries · agencies · defense · corporates

from €90k + services

- All 8+ domains, bespoke indices (Index Crafter)
- Dedicated agents & SLA
- Full APIs + enterprise integration
- On-prem / EU-sovereign option
- Dedicated team + co-design

Access · Enterprise

Subscription (KaaS)

annual, tier-based license

Pay-per-use API

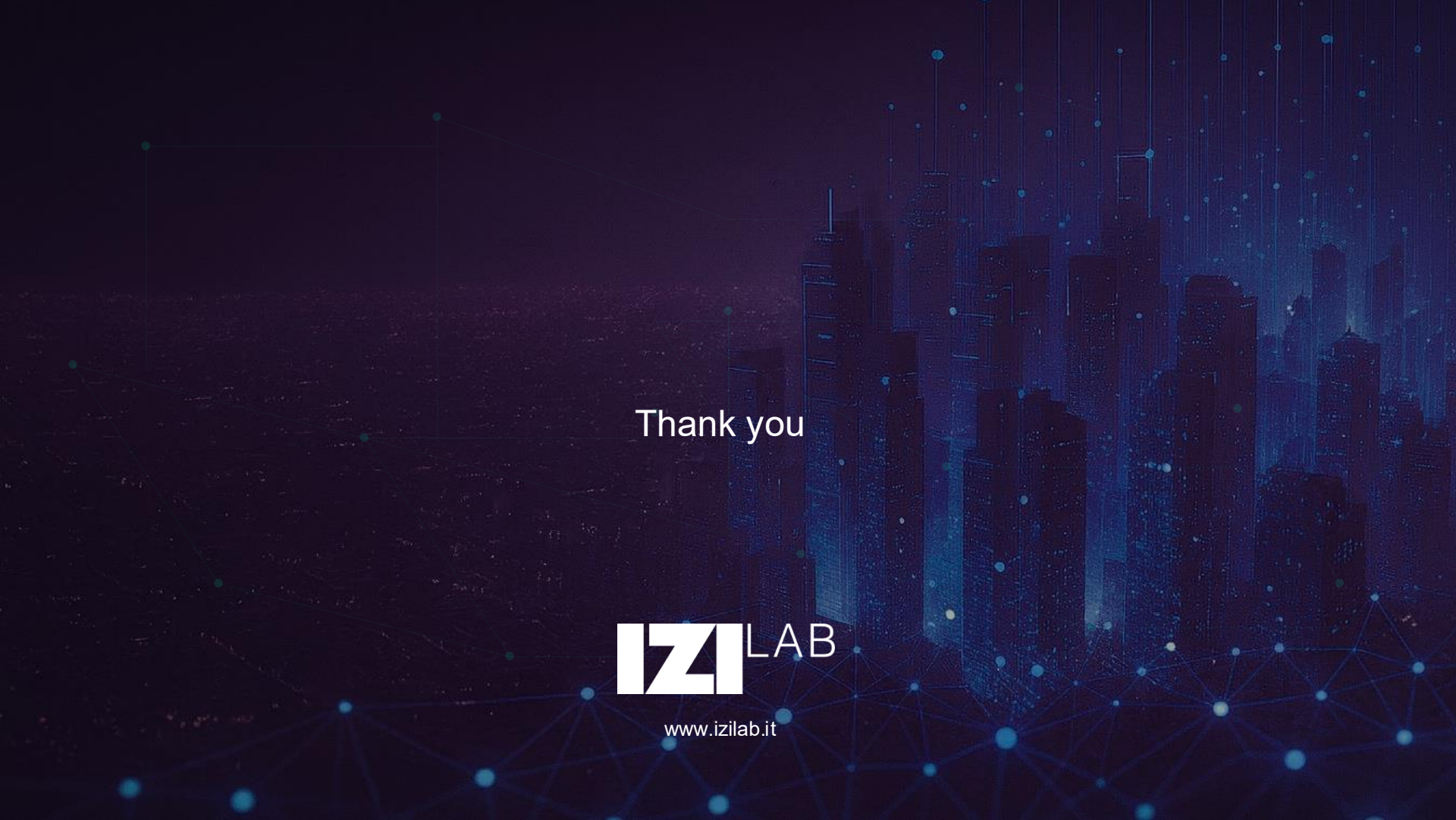
metered marketplace calls

Co-design services

bespoke indices, academic validation

Chorema

From the Greek χώρα (chōra) — space, territory, place, a chorema is an elementary graphic model representing the fundamental structures of geographic space organization. It is, in essence, the atom of analytical cartography: the minimal form through which the structure of a territory can be described.



Thank you



www.izilab.it